Adaptive Material Matching for Hyperspectral Imagery Destriping

Jia Li, Junjie Zhang, Fansheng Chen, Kai Zhao, and Dan Zeng

***Abstract*— Due to instrument instability, slit contamination, and light interference, hyperspectral images often suffer from striping artifacts, which greatly impairs the data quality. Real hyperspectral data are usually characterized by a small amount of historical data, complex material distribution, insignificant periodicity of noise, and so on, which brings significant challenges for the destriping task. However, the assumptions made by traditional destriping methods are often inconsistent with these characteristics. To this end, we propose a novel destriping method based on adaptive material matching (MAM) without making explicit assumptions of hyperspectral data. Specifically, to identify pixels that belong to the same material, we propose a principal material analysis (PMA) to adaptively generate thresholds within each superpixel. The pixels are matched by thresholding their vertical gradients and leveraging both inner stripe gradient feature (ISGF) and neighbor-stripe geometry feature (NSGF). Correction pixels selected from the same material can then be used to calculate the offsets and gains of pixels to adjust adjacent columns. To further improve the stability of the destriping process, we generate a set of correction candidates for each column and select the optimal candidate by considering the prior distribution and destriping nonuniformity. The stripe noise within the whole image is finally removed by iteratively performing the correction between adjacent columns. We compare the proposed model against traditional and deep learning methods on both synthetic and real hyperspectral images. The promising results indicate that MAM can effectively remove the image stripes, retain original image information, and improve the nonuniformity.**

***Index Terms*— Destriping, hyperspectral image, material matching (MAM), nonuniformity correction.**

I. Introduction

AUTOMATICALLY understanding and interpreting hyperspectral imagery is critical in various remote sensing applications, such as ground object classification [1], [2], endmember extraction [3], unmixing [4], [5], and target detection [6]. However, the interference of stray light, slit pollution, and instability of imaging instruments often cause stripe noise and nonuniformity of hyperspectral images during the acquisition. In particular, a pushbroom sensor produces a significant amount of stripe noise in hyperspectral images, which may severely degrade the visual quality [7]–[10] and affect the subsequent tasks. Therefore, it is essential to perform destriping and nonuniformity correction for the hyperspectral images acquired by these sensors [11]–[13].

In the past few decades, a variety of methods dedicated to hyperspectral image destriping have been proposed to obtain high-quality images. Based on the stripe removal principle, the existing methods can be roughly grouped into four categories [14], i.e., filtering-based [10], [12], [15]–[21], statistical-based [19], [22]–[27], optimization-based [7], [11], [28]–[54], and deep learning-based [55]–[65], each of which possess their unique advantages as well as limitations. Filtering-based methods remove the stripe noise by designing an appropriate filter according to the stripe noise characteristics in both spatial and transformed domains, while the statistical-based methods assume that the signal acquired by each pixel in the detector of pushbroom imaging system shares the same statistical distribution, among which histogram matching [22], [23] and moment matching (MM) [24], [25] are representative methods. On the other hand, the optimization-based methods regard the removal issue as an ill-posed inverse problem. The clean image can be well estimated from the striped image via some constraint terms, such as fidelity term, TV model, low rank, and sparsity. The fidelity term retains the information of the original image to the greatest extent, and the TV model makes the estimated clean image smoother. Low rank and sparsity are set for the image spectrum or noise components so that the algorithm can adequately separate the stripe. Most recently, some deep learning methods have been proposed. These methods learn the feature of the stripe or image in a supervised manner to remove the stripe.

However, real hyperspectral data are usually characterized by a small amount of historical data, complex material distribution, very low periodicity of noise, and so on. The assumptions made by traditional methods are often inconsistent with these characteristics, such as the assumption of material distribution in statistical-based methods and the dependence of some filtering-based methods on the periodicity of noise. To address these concerns, as shown in Fig. 1, we propose a hyperspectral image destriping method based on adaptive material matching (MAM) without explicit assumptions regarding the hyperspectral data, which locates the same material pixels within a local region to correct images.

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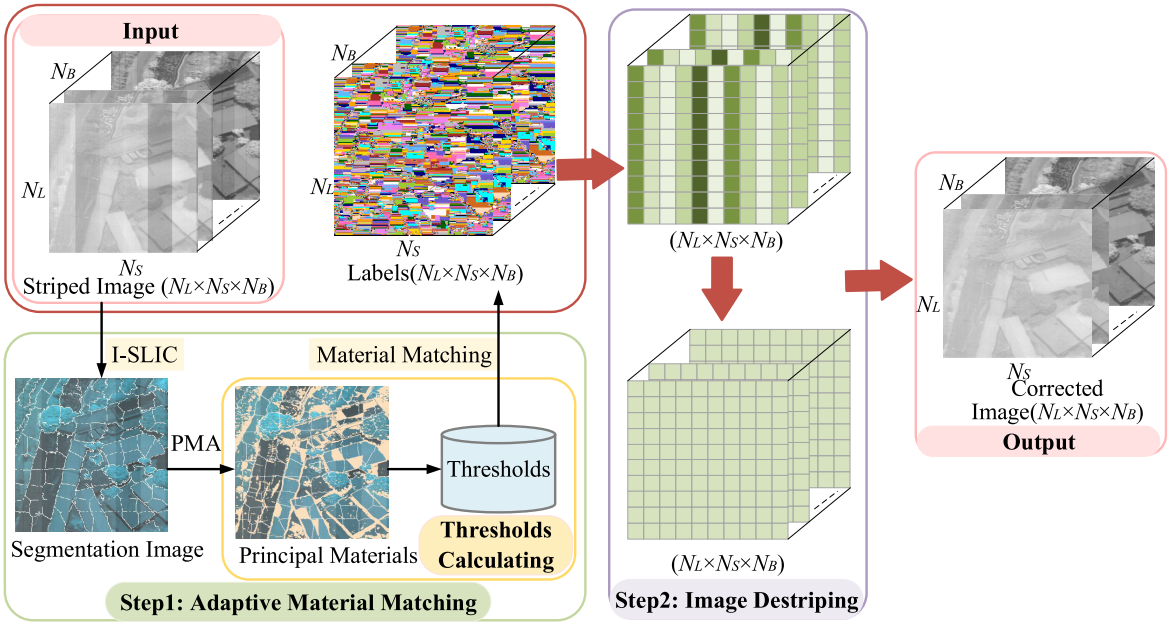


Fig. 1. Flowchart of the hyperspectral image destriping procedure with MAM. First, the hyperspectral image is segmented by I-SLIC. In each superpixel, the principal material pixels are obtained through principal material analysis (PMA). The thresholds are then calculated according to the principal material pixels, and each pixel is assigned with the material label using thresholds and the predefined feature distance. Finally, the stripe noise is removed using an improved two-point algorithm based on the correction pixels selected from the matching results.

More specifically, in a given scene, though horizontal gradients are compromised by stripes, the vertical gradients of pixels are usually consistent when pixels belong to the same material and possess close spatial positions. Therefore, we propose to segment the entire image into superpixels and perform a PMA to adaptively generate thresholds within each superpixel based on the vertical gradients of principal material pixel. With generated thresholds, we screen out the candidate pixels that potentially belong to the same material, and then, a predefined feature distance measurement is carried out to match the same material pixels within each superpixel, of which the inner stripe gradient feature (ISGF) considers the vertical gradient consistency, while the neighbor-stripe geometry feature (NSGF) considers the constraint of spatial distance.

With the same material pixels identified, the appropriate correction pixels that belong to two materials are selected for an improved two-point correction to calibrate adjacent columns. A column is first selected as a reference column, and then, the image is iteratively corrected column by column, centered from the reference to the most left and right columns. To further improve the robustness of the correction process, multiple groups of correction pixels are selected to generate a set of correction candidates for each column. We consider both prior distribution and destriping nonuniformity to select the final result, and namely, the optimal destriping has a similar distribution of vertical gradients with the original image and also is close to its adjacent reference column. By selecting suitable pixels of the same material in a local area to correct the image, there are no explicit assumptions about the data distribution and noise. Overall, our main contributions are summarized as follows.

1) We propose an adaptive MAM method to remove the stripe noise of hyperspectral images based on the PMA and predefined stripe feature distance, which is free from the explicit assumptions of hyperspectral data.
2) To improve the robustness of destriping process, we propose to generate the correction candidates and select the optimal solution based on the image uniformity as well as retaining the original image information in the correction process.
3) Compared to traditional and deep learning methods, our proposed method achieves the best results on both synthetic and real data. Detailed investigations are conducted to verify the effectiveness of the proposed adaptive matching mechanism and candidate selection.

The remainder of this article is organized as follows. Section II introduces the existing related methods. The proposed method is described in detail in Section III. To verify the effectiveness of the proposed method, experimental results are presented in Section IV. Furthermore, the image quality and information preservation are analyzed quantitatively in Section V. Section VI concludes this article.

II. Related Work

We coarsely group the related destriping methods into four categories, namely, filtering-based, statistical-based, optimization-based, and deep learning methods, and we introduce them in Section II-A.

A. Filtering-Based Methods

Filtering-based destriping methods aim to separate clean signals from the noisy ones mainly through two types of

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filtering operation, i.e., spatial domain and domain transform filters. Crippen *et al*. [15] proposed a spatial domain low-pass filter to remove the stripe, while the domain transform filter includes the Fourier transform and Wavelet decomposition, of which the Fourier transform methods rely on the assumption that stripe noise shows strong periodicity in the spatial domain [16], [17], [19]. However, noise components and useful signals are difficult to be accurately separated by the filter in the frequency domain. The removal of stripe noise often accompanies the loss of useful signals. Furthermore, ringing artifacts may arise at the points where the input radiances change abruptly [36]. On the other hand, Wavelet decomposition is an efficient denoising and destriping technique for remote sensing images [26], [27]. The image is transformed into a wavelet domain and scaling directional properties are applied to remove stripe noise [10], [20], [21]. Nonetheless, the effectiveness of Wavelet decomposition-based methods is affected by the selection of the wavelet transform function [7], [12].

B. Statistical-Based Methods

Statistical-based methods, such as histogram matching [22], [23] and MM [24], [25], make the assumption that the signal acquired by each pixel in the detector of pushbroom imaging system shares the same statistical distribution. Assuming that there is a linear relationship between the response values of detector pixels and the input radiances and the image scene is evenly distributed, the baseline objects within each column of pixels in the scanning direction will have similar radiation distribution. In order to eliminate the stripe noise, histogram matching adjusts the histogram of each detector to the same desired histogram, such as the histogram of the whole image. Similarly, Gadallah *et al*. [24] proposed the MM, which assumes that the mean and standard deviation of all detectors are approximately equal. Therefore, stripe noise is removed by adjusting the mean and standard deviation of each detector to those reference values based on the entire image. Statistical-based methods are straightforward and efficient to implement, yet the assumptions for data distribution limit the effectiveness of stripe removal, namely, when the distribution of ground objects is complex, the features acquired by each detector pixel will change greatly. In this case, it is not consistent with the assumption of statistical distribution similarity, which leads to unsatisfactory results. In summary, the performance of statistical-based methods strongly depends on the statistical distribution of data for each detector.

C. Optimization-Based Methods

Optimization-based methods can be divided into two categories: variation-based methods [11], [28]–[35] and low-rank-based methods. The former methods generally remove stripe noise by optimizing the variational model. Shen and Zhang [28] first proposed an algorithm based on maximum *a posteriori* (MAP) for remote sensing image destriping in 2009, with Huber–Markov-based variational model as the prior likelihood probability density function (pdf). However, these methods ignore the directional characteristic of stripe noise. Therefore, some methods are presented to overcome the drawback mentioned above by constraining the smoothness of the gradients in the cross-stripe direction and retaining the gradients in the along-stripe direction [30]–[34]. In addition, some methods for modeling stripe prior have also been proposed and achieved excellent performance [29], [34].

The low-rank-based methods include priors for low-rank constraints on the data, such as images, abundance matrices, and stripe noise. According to the form of data processing, these methods can be divided into matrix-based [7], [36]–[45] and tensor-based [46]–[52] methods. The matrix-based methods take advantage of the low-rank property along the spectral mode by lexicographically ordering the 3-D cubic into a 2-D matrix [62]. For example, Acito *et al*. [36] estimated the noise components, including stripe noise and random noise, by projecting the image into the orthogonal subspace. However, the matricization techniques preliminarily vectorize all the image bands at the cost of losing spatial structures of the image cube [52]. Therefore, some methods for directly modeling tensors are proposed, which could more faithfully deliver the underlying information of the hyperspectral image cube without destroying the spatial structures.

D. Deep Convolutional Neural Network Methods

With the success of deep learning on image tasks [66]–[69], deep convolutional neural network (CNN)-based methods dedicated to hyperspectral image destriping have also been proposed [55]–[65]. These methods allow the CNN to learn the ability to remove stripe through a large number of training samples. The original CNN-based method directly outputs a clean image after multilayer convolution [55]–[57]. Some subsequent algorithms introduce the residual learning into CNN, which employ the network to learn the stripe noise for separating clean images from stripe images [58]–[63], [65]. Moreover, the gradient features of the image are input to the network as additional information and so on [63], [65]. Zhang *et al*. [65] presented the spatial–spectral gradient network (SSGN) for removing mixed noise in hyperspectral images. In addition, Chang *et al*. [62] designed the two-stream wavelet enhanced U-net (TSWEU) model to focus on the relationship between clean images and striped noise, which better facilitates distinguishing them from each other. Although deep learning methods have shown promising results for the task of hyperspectral images destriping, the existing CNN-based methods generate the training samples by adding stripe noise to clean images, which is different from the real stripe images.

III. METHODOLOGY

A. Overview

To tackle hyperspectral image destriping without making the explicit assumption of data distribution, we propose a two-stage solution, i.e., adaptive MAM and destriping. The overall process is shown in Fig. 1.

For adaptive MAM, we segment the original image into superpixels by the improved simple linear iterative clustering technique (I-SLIC) [70]. In each superpixel, the principal material pixels are determined by the PMA based on the selected band information. The thresholds are calculated

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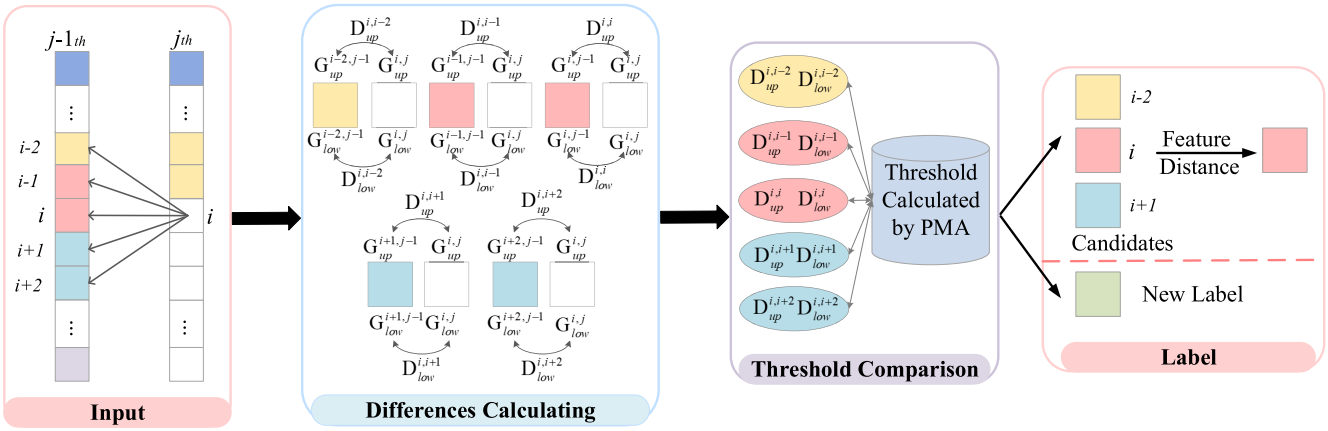


Fig. 2. Illustration of adaptive MAM for one pixel. First, we calculate the upper and lower gradients (G_{up} and G_{low}) concerning the target pixel and five labeled neighborhood pixels. Then, the upper and lower gradient differences (D_{up} and D_{low}) are compared with the threshold to determine candidate pixels. If the candidate pixels exist, the label is selected from them according to the predefined feature distance. If there are no candidate pixels, a new label is assigned to the target pixel.

according to the principal pixels in each superpixel of each band. We select the candidate pixels for each pixel by thresolding the differences of vertical gradients. The final material label of each pixel is determined from the candidate pixels based on the predefined feature distance combining the ISGF and NSGF.

After we assign the material label to each pixel, an improved two-point algorithm is utilized to correct the image column by column using the correction pixels selected from the matching results. To improve the robustness of the destriping process, we generate multiple correction candidates for each column by cross-checking multiple groups of correction pixels, and we select the optimal correction column based on the image uniformity as well as to retain the original image information. Next, we describe each proposed stage in detail.

B. Adaptive MAM

The vertical gradients of pixels located in the adjacent column, including the upper and lower gradients, are computed for MAM. If the vertical gradients of two pixels are close enough, they can be considered as the same material. Therefore, we set a threshold in each superpixel to limit their differences. The details of calculating the threshold will be given later. We determine whether two pixels located in the adjacent columns belong to the same material by the thresholds we set, as well as the predefined feature distance based on the ISGF and NSGF.

First of all, the I-SLIC algorithm [70] is employed to segment the image. Due to the interference of stripe noise and the complex contents of hyperspectral image, the resulted superpixels are rather coarse, namely, each superpixel often contains multiple materials. Since we set a threshold in each superpixel of each band based on the vertical gradients of principal material pixels, if the differences of vertical gradients between two pixels exceed the threshold, they are considered to be different materials. During the matching process, pixels belonging to the same material are given the same label. Each pixel of the first column in the superpixel is allocated with a different label beginning with 1(one). Labels are then iteratively determined column by column from left to right.

Assuming that the labels of pixels in column $(j-1)$th have been allocated, we show an example of label assignment for ith pixel in the jth column noted as the target pixel in Fig. 2. We first define five pixels indexed from $(i-2)$th to $(i+2)$th in the adjacent $(j-1)$th column as the neighborhood of the target pixel and compute their upper and lower gradients, respectively, as follows:

$$\begin{aligned} G_{\text{up}} &= \text{DN}_{i-1} - \text{DN}_i \\ G_{\text{low}} &= \text{DN}_i - \text{DN}_{i+1} \end{aligned} \tag{1}$$

where DN_i, DN_{i-1} and DN_{i+1} represent the detector response digital numbers (DNs) of the current pixel, the upper pixel, and the lower pixel in the same column, respectively. We only compare against five neighbors in the adjacent column since the pixels with close spatial distances have high probabilities of belonging to the same material. Second, the differences between the upper and lower gradients among the target pixel and its five neighbors are, respectively, calculated by

$$\begin{aligned} D_{\text{up}} &= G_{\text{up}_t} - G_{\text{up}_n} \\ D_{\text{low}} &= G_{\text{low}_t} - G_{\text{low}_n} \end{aligned} \tag{2}$$

where G_{up_t} and G_{up_n} stand for G_{up} of the target pixel and its neighbors, respectively, and G_{low_t} and G_{low_n} stand for G_{low} of corresponding pixels, respectively. By judging whether the differences of D_{up} and D_{low} are within the threshold of given superpixel, we can determine whether two pixels belong to the same material. If the difference is below the threshold, the neighbor pixel is considered as a candidate pixel, and if no candidate pixels are identified, a new label will be assigned to the target pixel.

In the case where more than one candidate pixel exists, the label of the target pixel is selected based on the stripe feature distance. We consider two types of stripe features from both vertical gradient and spatial position perspectives: one is NSGF, which is computed as the sum of D_{up} and D_{low};

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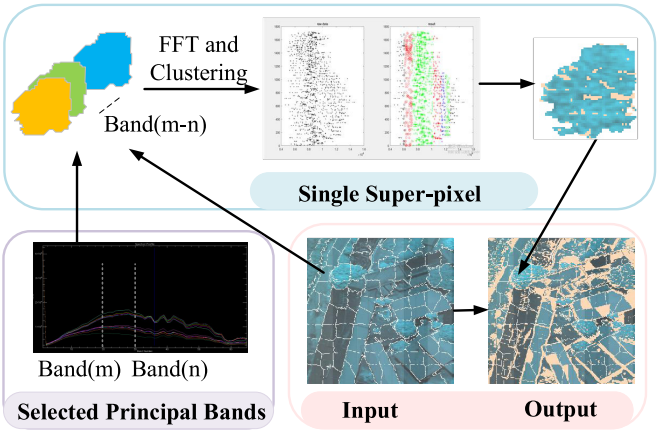


Fig. 3. Illustration of PMA. FFT is performed on each pixel based on the spectral information in the selected bands, and then, the values of the dc component are clustered into several groups. The pixels that belong to the largest cluster are employed as the main material pixels.

the other is ISGF noted as d. It is the longitudinal geometric distance between two pixels. The final feature distance is defined as follows:

$$D = \frac{1}{1+e^{-d}} \times (D_{\text{up}} + D_{\text{low}}). \tag{3}$$

The label of the candidate pixel with a minimum feature distance is therefore assigned to the target pixel. With the labels of all pixels allocated, they will be further used for destriping.

Thresholds: Here, we present the details of setting thresholds. In each superpixel, we select principal pixels through PMA to calculate thresholds. The whole process is divided into two steps, as shown in Fig. 3.

For the first step, a portion of the continuous bands with a large difference in spectral characteristics of different materials is filtered from all bands. Specifically, the variance of each band is computed and sorted in a descending order. The first fifth of the sorted bands is taken out with their serial numbers retained. Then, we divide these bands into several groups by applying a box plot based on the serial number. The bands in the largest group are finally selected as the principal bands.

Next, the spectral information of each pixel in the selected principal bands is transformed by the fast Fourier transformation (FFT). The values of the dc component (the constant component of the Fourier transform) after the FFT are clustered into several groups, of which pixels that belong to the largest cluster are selected as the principal material within the given superpixel. Fig. 3 (bottom right) shows the schematic after the main material pixels are selected, and the light yellow parts are the abandoned pixels.

The computation of thresholds based on the principal material pixels is shown in Fig. 4. The calculation of the thresholds consists of three steps. First of all, we calculate the vertical gradients ($G_1, 1, G_1, 2, G_1, 3, \ldots, G_n, 6$ in Fig. 4) between adjacent principle pixels on each column. Second, the differences of these vertical gradients located in two adjacent columns are calculated. Finally, we sort the absolute values of all differences and select the median value as the threshold of

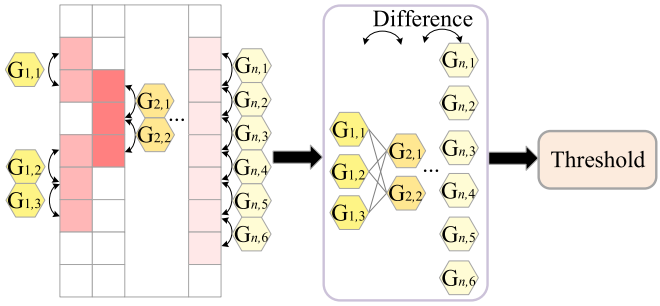


Fig. 4. Illustration of threshold calculating. The gradients $(G_{1,1}, G_{1,2}, G_{1,3}, \ldots, G_{n,6})$ between adjacent main material pixels are calculated on each column. The gradient differences between adjacent columns are then calculated. The median of all gradient differences is selected as the threshold of the superpixel.

one superpixel. All thresholds are obtained by operating the above procedure in each superpixel.

C. Image Destriping

Based on the MAM results, an improved two-point correction method is adopted to remove stripe noise. We first select 1-pixel column from the original image as the baseline column for each band. This column does not need to be corrected. To ensure that the corrected image can retain the original information to a maximum extent, the DNs of all pixels are summed in each column, and the column with the median value of the summation is considered as the baseline. This baseline is then taken as the starting position, and we iteratively perform the destriping in both left and right directions. A more detailed pseudocode is shown in Algorithm 1.

Taking the left direction correction as an example, assuming that the baseline is located in the jth column, the baseline is considered as the reference column, and the $(j$-1)th column is the column to be corrected. We employ the two-point correction method to destripe the $(j$-1)th column. After the correction is completed, the resulted column $(j$-1)th is taken as the reference column, and the $(j$-2)th column is to be corrected. The operation is repeated until the left part of the image is fully corrected. The right part of the baseline column is corrected in the same way.

For the simple two-point correction [71], [72], given two pairs of the same material pixels from two different superpixels, a group of correction pairs (four pixels) is set as baseline. Each pair is composed of one pixel from the reference column and another one from the uncorrected column. This group of pixels is further used to obtain the correction coefficients k and b based on the linear regression

$$\begin{aligned} k &= \frac{\text{V}_{\text{ref}}(\phi_1) - \text{V}_{\text{ref}}(\phi_2)}{\text{V}_{\text{un}}(\phi_1) - \text{V}_{\text{un}}(\phi_2)} \\ b &= \text{V}_{\text{ref}}(\phi_1) - k \times \text{V}_{\text{un}}(\phi_1) \end{aligned} \tag{4}$$

where $\text{V}_{\text{ref}}(\phi_1)$, $\text{V}_{\text{ref}}(\phi_2)$, $\text{V}_{\text{un}}(\phi_1)$, and $\text{V}_{\text{un}}(\phi_2)$ are the DNs of the correction pixels of the reference column and the uncorrected column, respectively. The response DN $\text{V}_{\text{un}}(\phi)$ of each pixel is corrected to $\text{V}'(\phi)$ using k and b

$$\text{V}'(\phi) = k \times \text{V}_{\text{un}}(\phi) + b. \tag{5}$$

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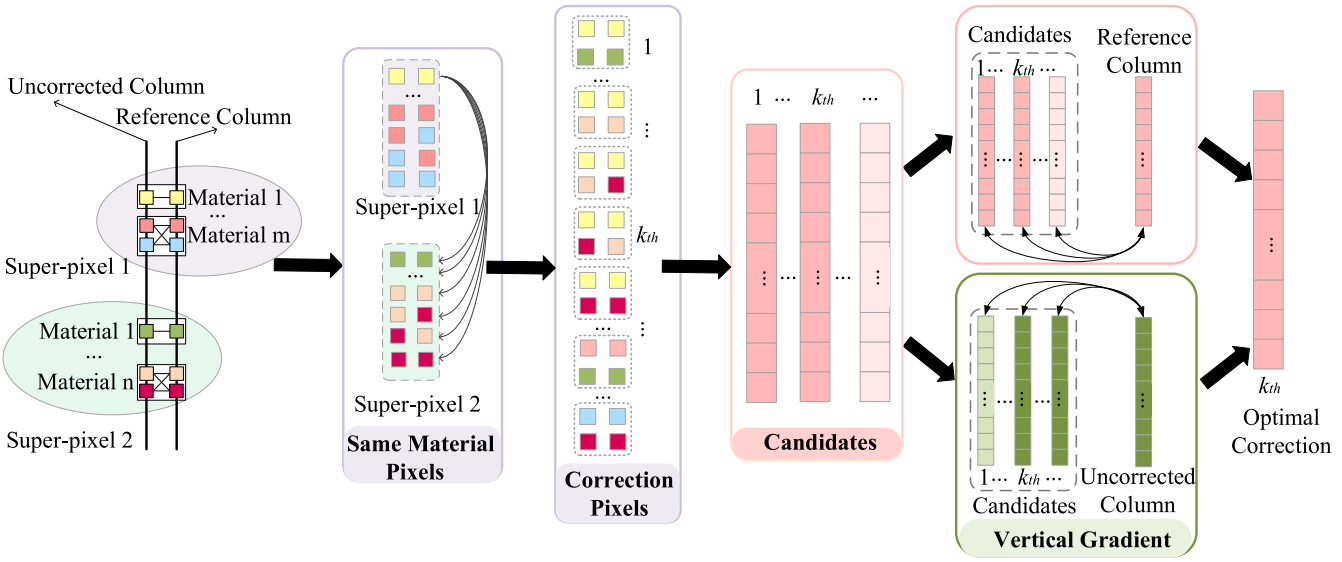


Fig. 5. Illustration of the correction procedure for one column. Based on the matching results, multiple groups of correction pixels are selected from the two superpixels to obtain multiple correction column candidates. The optimal correction column is then selected from the candidates according to the vertical gradient difference and horizontal same material gradient.

To improve the simple two-point correction with our proposed adaptive MAM, we propose to generate multiple groups of correction pairs to enhance the robustness of the destriping process. The improved two-point correction of adjacent columns is shown in Fig. 5. More specifically, considering the unstable performance of the linear regression when one group of correction pairs is randomly selected, we generate multiple groups as inputs. Namely, two superpixels with the largest mean difference of DNs are selected from all superpixels crossed by the two columns to ensure a large material difference. In this way, the error in estimating the correction coefficients can be minimized. All of the same material pixels in the selected two superpixels are combined into correction pairs, e.g. m pixel pairs from the first superpixel and n pairs from the second one will generate $m \times n$ sets of correction groups in total, and we then obtain $m \times n$ candidates by following the two-point correction.

We select the optimal correction column from $m \times n$ candidates by considering both image uniformity as well as retaining the original image information in the correction process, i.e., on the one hand, in the same MAM region, there only exists a small difference between the corrected column and the adjacent reference column since the pixels of the same material after correction should present approximately equal DNs. On the other hand, the difference between the vertical gradients of the column before and after the correction should also be small, ensuring that the corrected column retains the original image information to the largest extent. Therefore, we monitor the vertical gradient difference and horizontal same material gradient to evaluate the candidates, i.e.,

$$D_{\text{ver}} = \sum_{i=1}^{p-1} \left| G'_i - G_i \right|$$
$$D_{\text{hor}} = \sum_{j=1}^{q} \left| M'_j - M_j \right| \tag{6}$$

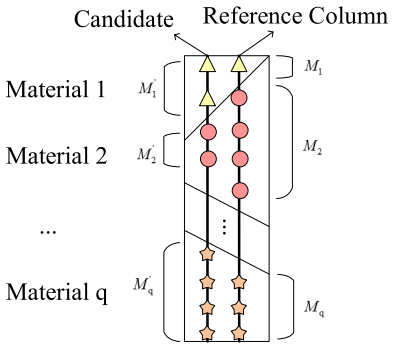


Fig. 6. Illustration of horizontal same material gradient calculation area. m' and m are the MVs of pixels in the same material region of the two column.

where G'_i and G_i are the ith vertical gradient of candidate and uncorrected column, respectively, p is the number of pixels in the uncorrected column, M'_j and M_j are the mean values (MVs) of pixels in the jth same material region of the reference column and uncorrected column, respectively, and q is the number of the whole material regions through which the two columns pass. The ith vertical gradient is obtained by subtracting the DN of the ith pixel from the $(i+1)$th pixel. Fig. 6 shows the parts of the two columns in different material regions. We first sort both D_{ver} and D_{hor} of all candidate corrections in an ascending order, and then, we iterate through the sorting list of D_{hor}. The ith candidate of D_{hor} is selected as the optimal solution, only when it is also located in the top i list of D_{ver}. In this way, the optimal correction is considered to possess both image uniformity and retain the original image information to the largest extent.

IV. Experimental Results and Analysis

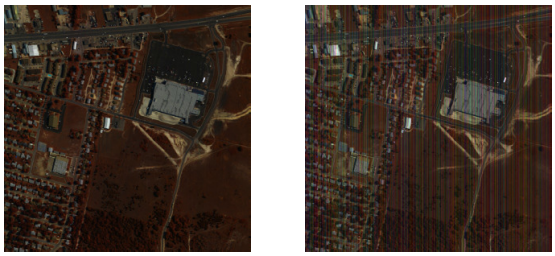
In this section, we compare our proposed MAM against five classic destriping methods, namely, MM [24], ASSTV [33], LRSID [41], ICSRN [57], and SSGN [65], on one synthetic image and three real images. We visualize and

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Algorithm 1 Destriping Process

Input: The hyperspectral image (size of $N_L N_S N_B$)
The material matching results (size of $N_L N_S N_B$);
Output: The corrected hyperspectral image (size of $N_L N_S N_B$);

for $b = 1 \rightarrow N_B$ **do**
  for $j = 1 \rightarrow N_S$ **do**
    $S_{bj} \leftarrow$ the sum of DNs of all pixels in the jth column of band b;
  end for
  $L \leftarrow$ the column corresponds to the median of S_{bj};
  $Num \leftarrow$ the serial number of L;
  $L_{\text{ref}} \leftarrow L$;
  for $j = Num \rightarrow 2$ **do**
    $L_{\text{un}} \leftarrow$ the j-1_{th} column in band b;
    Select the correction groups from the two super-pixels crossed by L_{ref} and L_{un};
    Compute the correction candidates according to Eq. (4) and (5);
    $L_{cor} \leftarrow$ the optimal corrected column according to Eq. (6);
    $L_{\text{ref}} \leftarrow L_{cor}$;
  end for
  $L_{\text{ref}} \leftarrow L$;
  for $j = Num \rightarrow (N_S - 1)$ **do**
    $L_{\text{un}} \leftarrow$ the j+1_{th} column in band b;
    Select the correction groups from the two super-pixels crossed by L_{ref} and L_{un};
    Compute the correction candidates based on Eq. (4) and (5);
    $L_{cor} \leftarrow$ the optimal corrected column based on Eq. (6);
    $L_{\text{ref}} \leftarrow L_{cor}$;
  end for
end for



(a) (b)

Fig. 7. False-color images of urban HYDICE. (a) Original image. (b) Striped image.

analyze the destriping effect of all compared methods as well as the proposed modules in MAM. Furthermore, we also give detailed quantitative analysis for the model performance.

A. Visualization of Destriping

1) Synthetic Image: In our experiments, the synthetic noisy image is generated by adding stripe noise to the clean hyperspectral image, of which the urban data from Hyperspectral Digital Imagery Collection Experiment (HYDICE) in October 1995 with 307 × 307 × 162 resolution is used as the clean image. Its spectral and spatial resolutions are 10 nm and 2 m, respectively. Fig. 7(a) shows a false-color composition of the original image. The generation of a noisy image can be formulated as: $V'_{l,j}(\varphi) = A_{l,j} \times \upsilon_{l,j}(\varphi) + C_{l,j}$, where $A_{l,j}$ and $C_{l,j}$ are the multiplicative and additive noise values and $A_{l,j}$ and $C_{l,j}$ are random values, but the same column shares the same $A_{l,j}$ and $C_{l,j}$. In the experiment, the ranges of $A_{l,j}$ and $C_{l,j}$ are set to [0.5, 1.5] and [10, 30], respectively. The generated noisy image is shown in Fig. 7(b). We show the correction results of bands 18 and 29 in Figs. 8 and 9, respectively.

TABLE I
MAIN FAHI PARAMETERS (VNIR, SWIR, CCD, AND MCT)

Item	VNIR	SWIR
Spectral range(μm)	0.4-0.95	0.95-2.5
FOV(○)	14.7	14.7
Detector/array size	CCD/1024×256	MCT/512×512
Spectral resolution(nm)	2.34	3
Band numbers	64	512

2) Real Images: Apart from synthetic data, to investigate the practicability of different methods, we also observe their effects on three real noisy images. The two images originated from the near-infrared (VNIR) and shortwave-infrared (SWIR) images produced by the full spectrum airborne hyperspectral imager (FAHI), which is the Chinese next-generation pushbroom hyperspectral image instrument [73]. Images with the sizes of 400 × 394 and 393 × 321 are cropped from VNIR and SWIR as the test images, where the main parameters are listed in Table I. Two false-color images are shown in Fig. 10(a) and (b). The last test image is MODIS image with 256 *times* 256 size, and its false-color image is shown in Fig. 10(c). The correction results of the VNIR image for bands 9 and 59, the SWIR image for bands 250 and 314, and MODIS for bands 4 and 6 are shown in Figs. 11–16.

3) Destriping Analysis: For both synthetic and real images, the visualization results indicate that the proposed MAM can effectively remove the stripe noise, especially for the noise of the real image, as shown in Figs. 11(a)–16(a), which contains continuous, discontinuous, and wide stripes. In particular, as shown in Fig. 11, only MAM can completely remove the stripe of such an image compared to other methods. The results in Figs. 11–16 indicate that the MAM technique is also suitable to be applied to low SNR images where the separation of target and noise signal is subtle.

For the comparisons against statistical-based methods, the classic MM method based on the statistical analysis achieves relatively good correction results on bands with the similar distributions of materials between columns, as shown in Figs. 12(b), 15(b), and 16(b). However, its performance is poor when the material difference between columns becomes large. For example, in Figs. 8(b), 9(b), 11(b), 13(b), and 14(b), the bright areas in the original image become darker after the correction. This is because the MM process requires a large amount of data to ensure that the MV and standard deviation of each column are approximately equal, and thus, its

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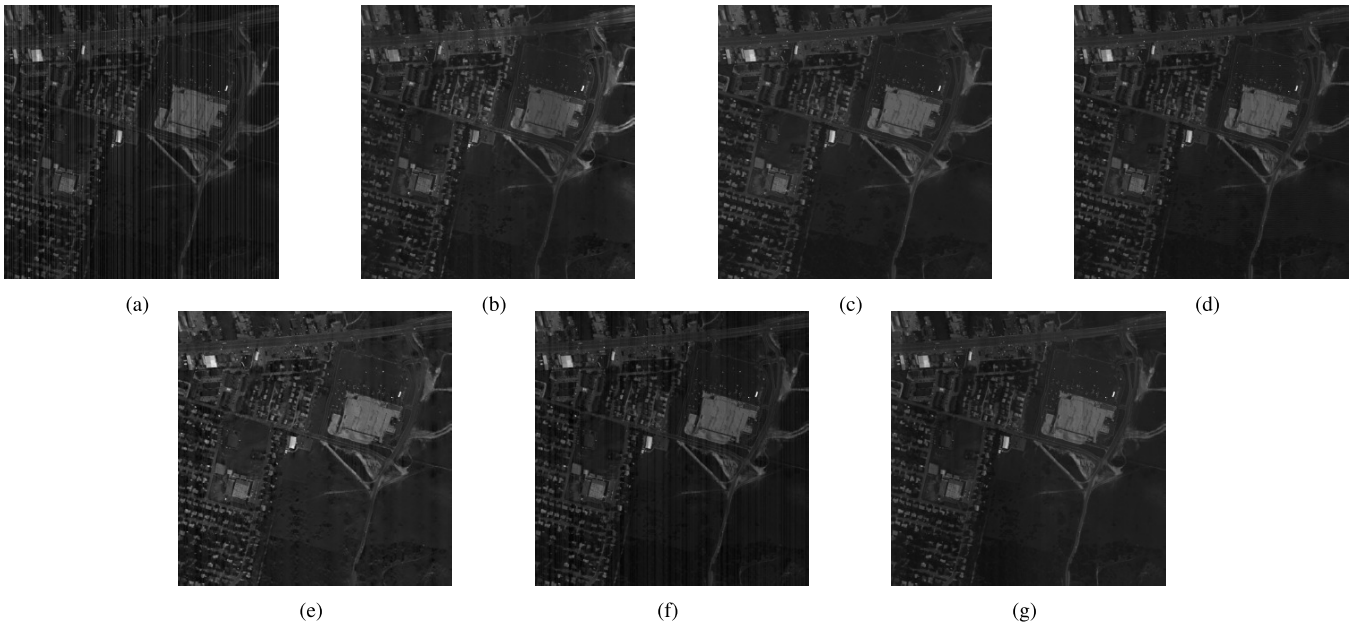


Fig. 8. Corrected urban HYDICE images for band 18. (a) Striped image. Images corrected by (b) MM, (c) ASSTV, (d) LRSID, (e) ICSRN, (f) SSGN, and (g) MAM.

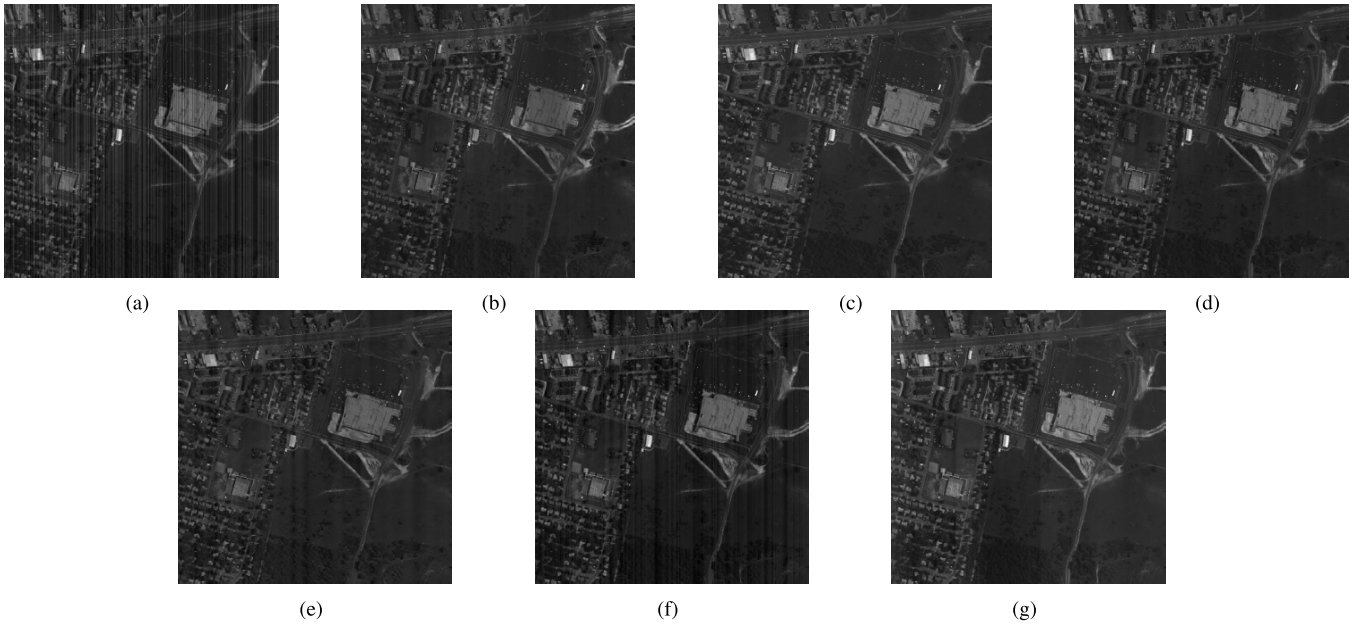


Fig. 9. Corrected urban HYDICE images for band 29. (a) Striped image. Images corrected by (b) MM, (c) ASSTV, (d) LRSID, (e) ICSRN, (f) SSGN, and (g) MAM.

performance becomes unstable when facing insufficient data or a large material difference between columns. Our proposed MAM process actively avoids this issue by selecting correction pixels from the MAM areas.

From Figs. 8(c) and (d), 9(c) and (d), 13(c) and (d), and 14(c) and (d), we can see that the optimization-based methods have a better stripe removal effect for thin stripes, whether it is a synthetic image or a real image. The ASSTV and LRSID methods obtain a more clear image with fewer stripes and a more visual smoothness appearance due to the effectiveness of the TV term on the image. The TV model aims to increase the smoothness between columns, so it has excellent performance for the stripe with large differences between adjacent columns. Unfortunately, they are not suitable for the wide stripe, especially in Figs. 11(c) and (d) and 12(c) and (d), and ASSTV and LRSID cannot completely remove stripe.

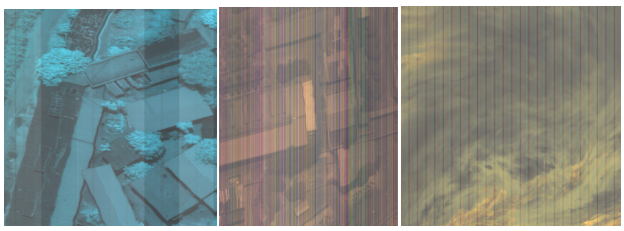


Fig. 10. False-color images of (a) VNIR, (b) SWIR, and (c) MODIS.

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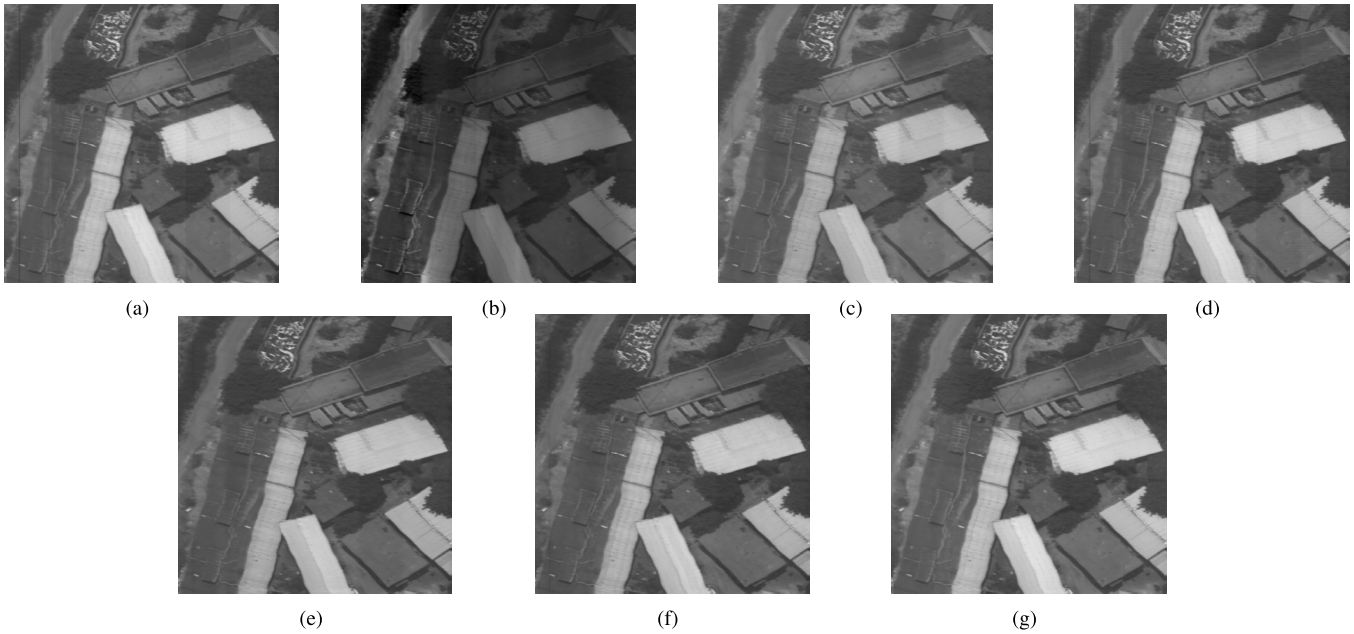


Fig. 11. Corrected VNIR images for band 9. (a) Original image. Images corrected by (b) MM, (c) ASSTV, (d) LRSID, (e) ICSRN, (f) SSGN, and (g) MAM.

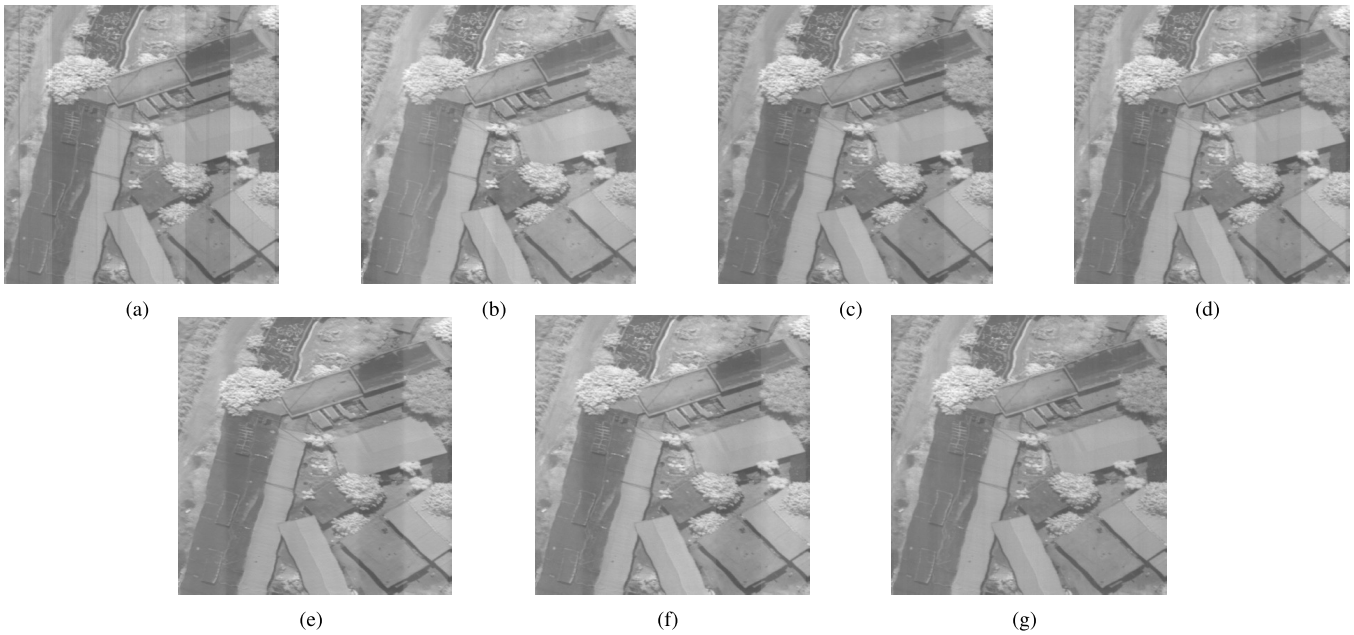


Fig. 12. Corrected VNIR images for band 59. (a) Original image. Images corrected by (b) MM, (c) ASSTV, (d) LRSID, (e) ICSRN, (f) SSGN, and (g) MAM.

As for the deep neural network model including ICSRN and SSGN, though they can be trained in an end-to-end fashion, we observe that it is more suitable for handling images with a light level of stripe noise, as shown in Fig. 11(e) and (f). However, its results are less satisfactory on images with severe stripes, as shown in Figs. 8(e) and (f), 9(e) and (f), 12(e) and (f)–15(e) and (f), and 16(e) and (f). Compared to the CNN, our carefully designed framework is more effective in removing the complex stripes.

B. Analysis of I-SLIC and Threshold

To further investigate the effectiveness of the proposed method, we present the results and analysis of the I-SLIC and different threshold setting methods of the matching process in this section.

In Fig. 17, we show the initial I-SLIC superpixel results, where Urban HYDICE, VNIR, SWIR, and MODIS images are segmented into 83, 96, 104, and 107 patches, respectively. It can be seen from the initial segmentation results that the I-SLIC performed poorly on the striped hyperspectral images. One of the main reasons is that the stripe noise seriously interferes with the spectral information of pixels in the images. Some of the longitudinal segmentation edges that appear in Fig. 17(c) and (d) follow the stripes rather than the true target boundaries of the ground object. Second, even with the light stripe interference, the ground scene in the hyperspectral

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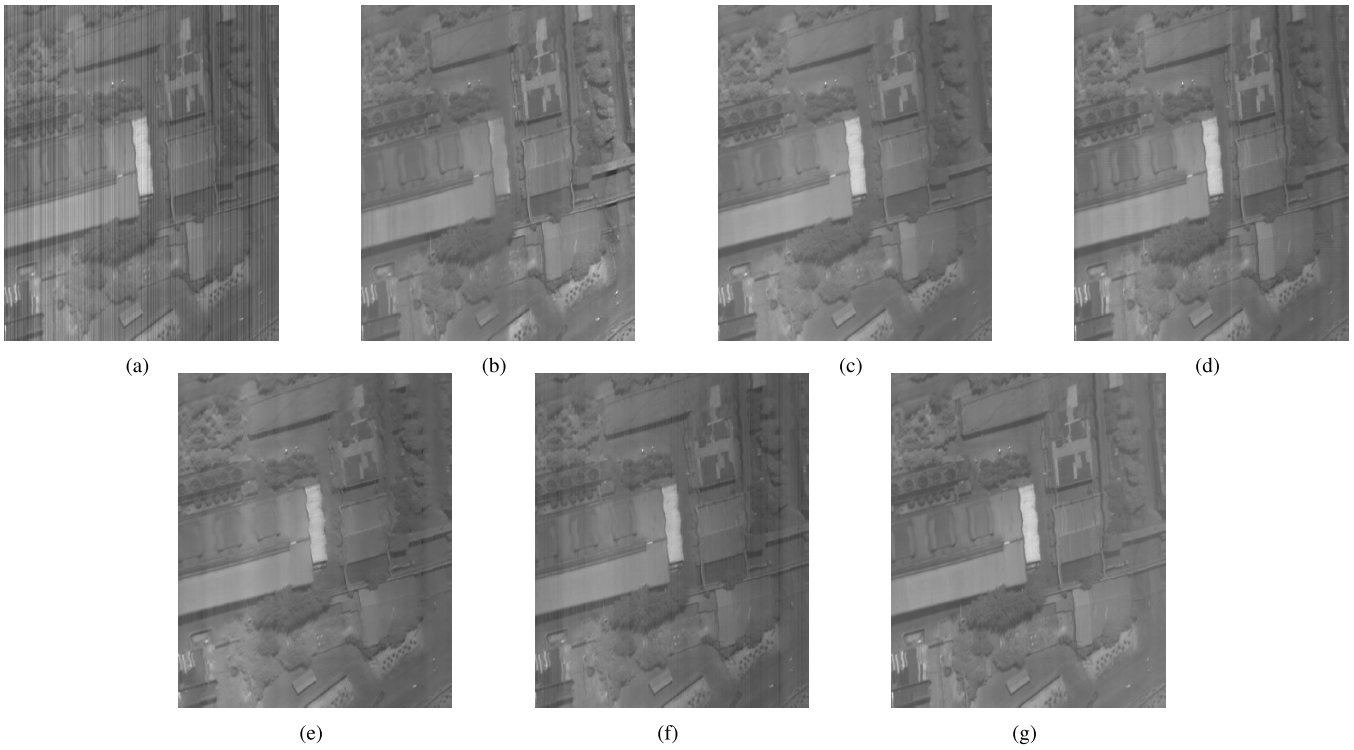


Fig. 13. Corrected SWIR images for band 250. (a) Original image. Images corrected by (b) MM, (c) ASSTV, (d) LRSID, (e) ICSRN, (f) SSGN, and (g) MAM.

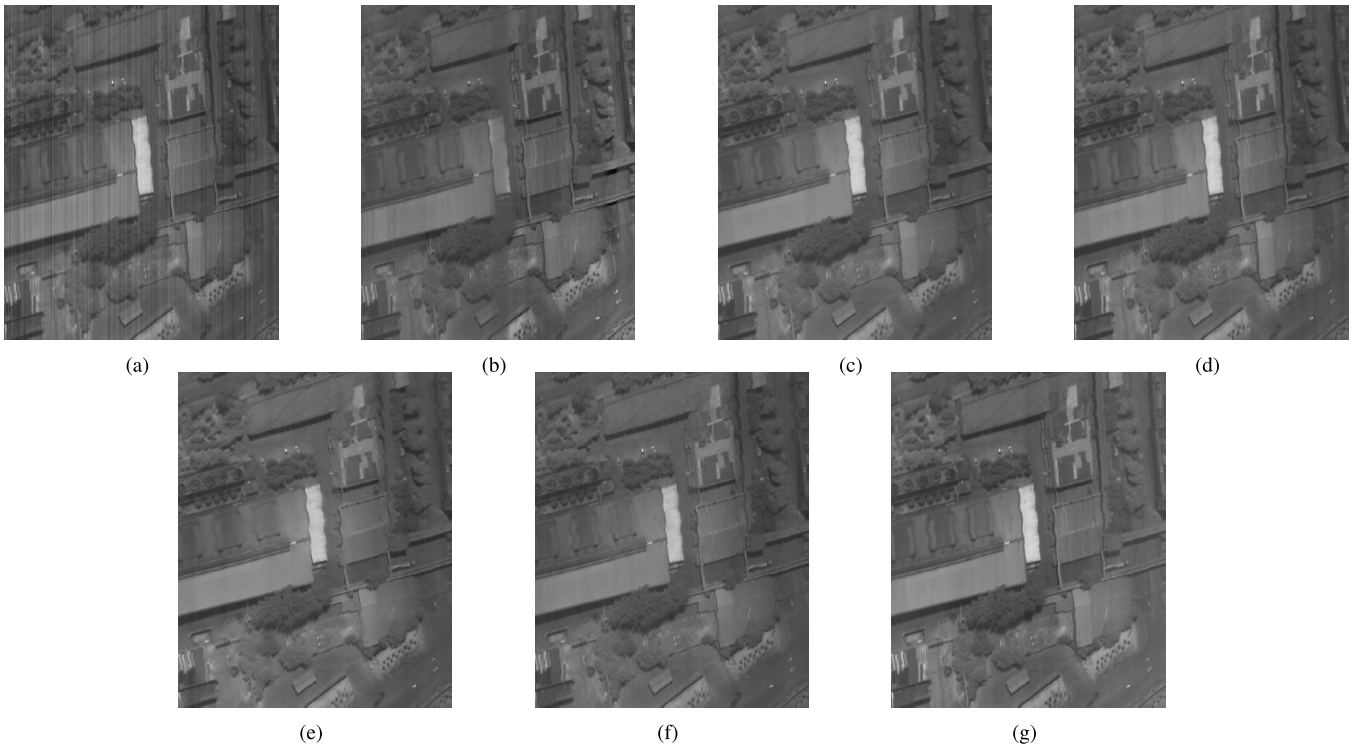


Fig. 14. Corrected SWIR images for band 314. (a) Original image. Images corrected by (b) MM, (c) ASSTV, (d) LRSID, (e) ICSRN, (f) SSGN, and (g) MAM.

images is rather complex, and the spatial resolutions of pixels are small, which may lead to large material differences in pixels with close spatial positions. As shown in Fig. 17(a), buildings and streets are adjacent and I-SLIC cannot accurately distinguish them. Therefore, it is necessary to match the materials in each superpixel to get more accurate same material pixels for further correction.

To demonstrate the effectiveness of the threshold setting method proposed in this article, we compare the results obtained by setting a random threshold for each band with our

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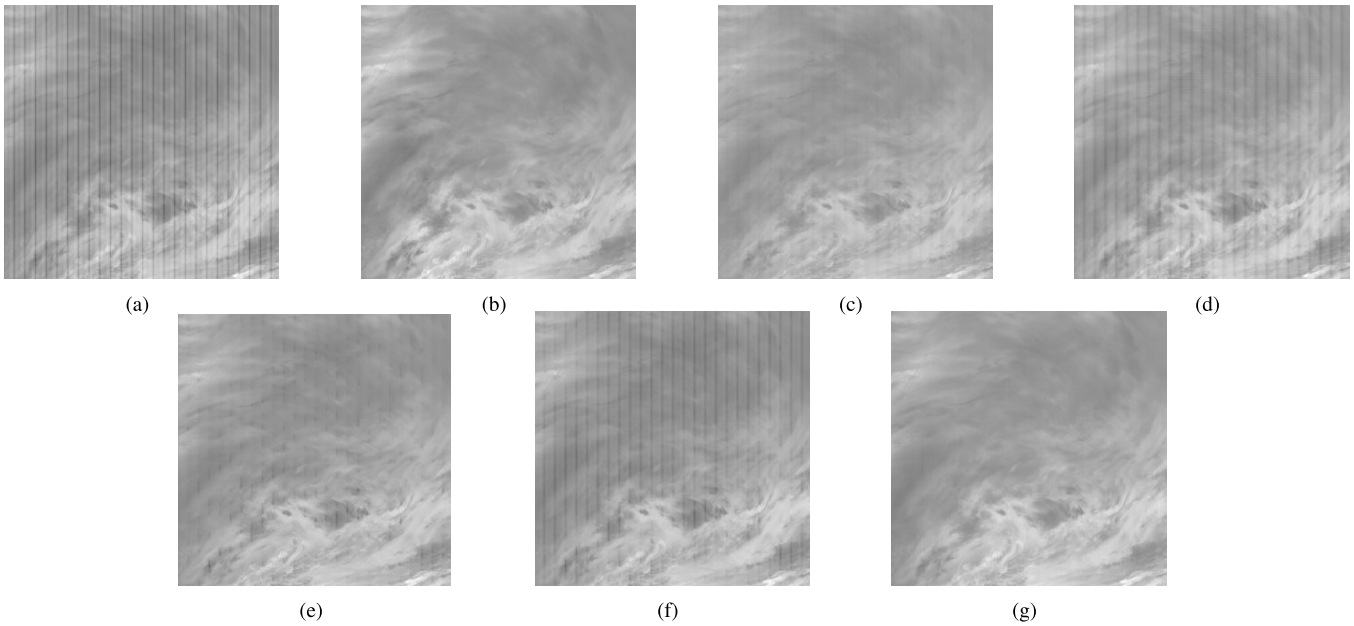


Fig. 15. Corrected MODIS images for band 4. (a) Original image. Images corrected by (b) MM, (c) ASSTV, (d) LRSID, (e) ICSRN, (f) SSGN, and (g) MAM.

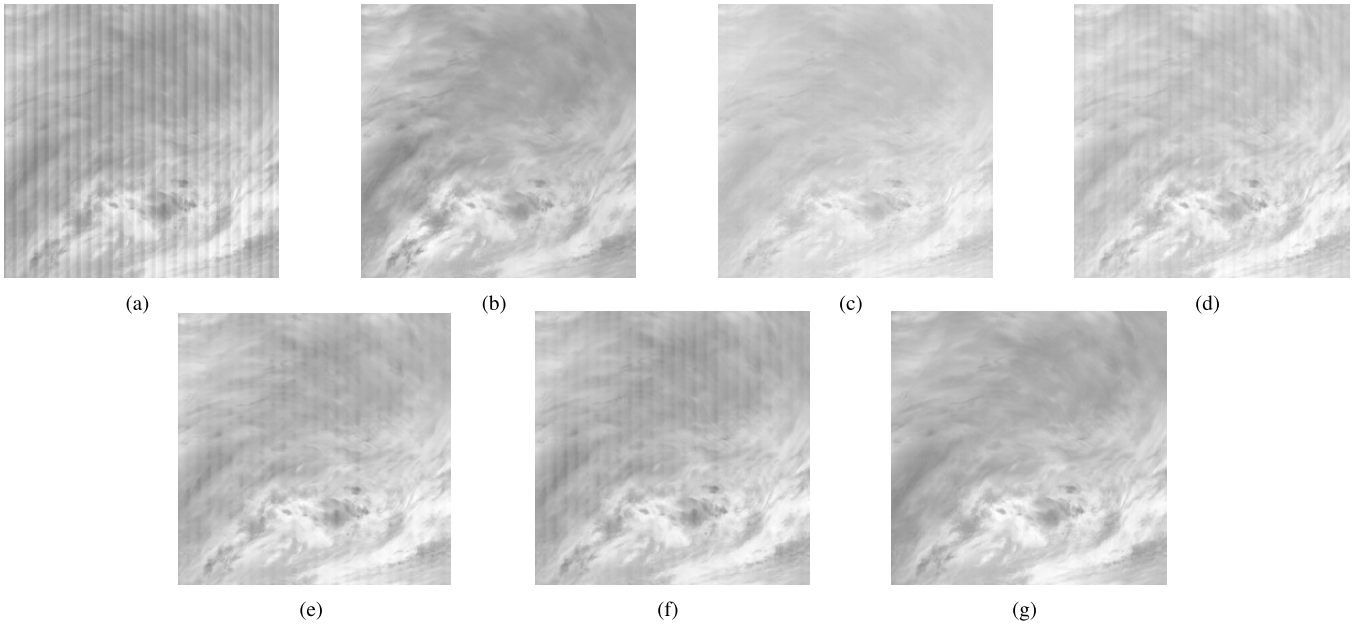


Fig. 16. Corrected MODIS images for band 6. (a) Original image. Images corrected by (b) MM, (c) ASSTV, (d) LRSID, (e) ICSRN, (f) SSGN, and (g) MAM.

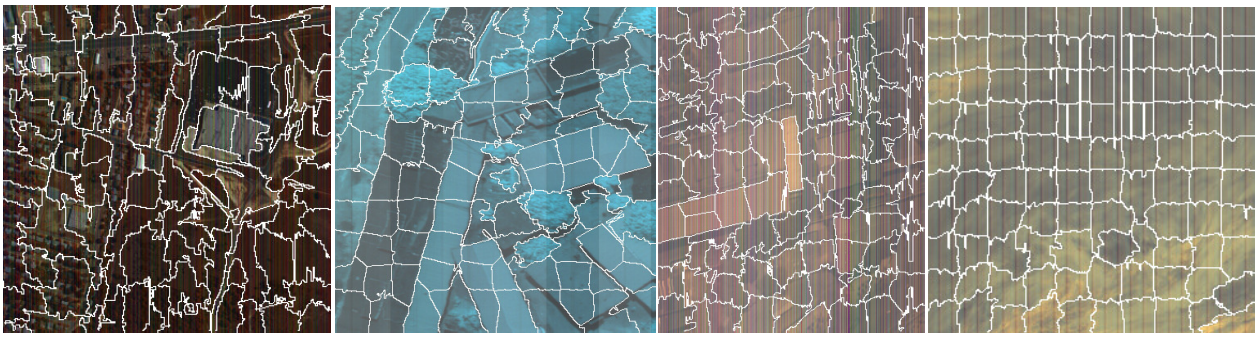


Fig. 17. Segmentation results of MAM for synthetic and real images. (a) Urban HYDICE image. (b) VNIR image. (c) SWIR image. (d) MODIS image.

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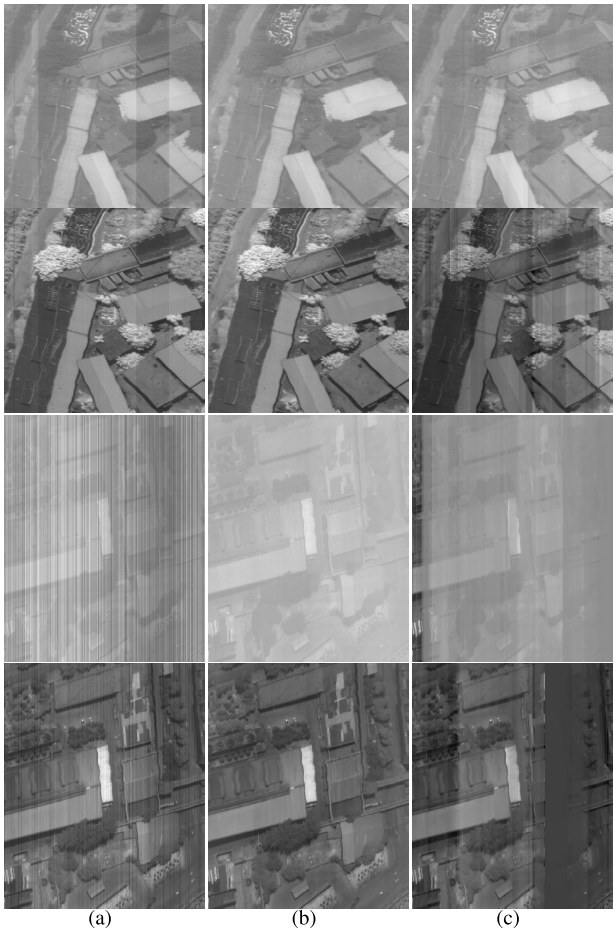


Fig. 18. Corrected images with different thresholds setting methods. Bands 3 and 54 of the VNIR image and bands 165 and 325 of the SWIR image are displayed sequentially from top to bottom. (a) Original image. (b) Corrected image obtained by MAM. (c) Corrected image obtained by random thresholds.

corrected image. The destriping results with different threshold setting methods for bands 3 and 54 of the VNIR image and bands 165 and 325 of the SWIR image are shown in Fig. 18. We set a random number as the threshold for the whole superpixel in one band, and the results are shown in Fig. 18(c). Fig. 18(a) and (b) shows the original and corrected image using our thresholds setting method, respectively. We can see that the random threshold method cannot completely remove the stripe. What is more serious is that an inappropriate threshold will increase stripe noise, causing the corrected image to have more stripe than the original image, as shown in the second row of Fig. 18(c). Inappropriate thresholds will affect the MAM, resulting in inaccurate material. During the calibration process, the two adjacent columns cannot get the same material. Therefore, the image is corrected inaccurately, which will also increase the nonuniformity of the image. We select the principle pixels through PMA to calculate the threshold for each superpixel, which can provide more accurate thresholds for correction to achieve excellent destriping performance, as shown in Fig. 18(b).

V. Validation

Apart from the visual observations, we also conduct quantitative analysis to verify our contributions.

A. Distribution of DNs

To demonstrate the effectiveness of stripe removal, we use line 150 in the MODIS image band 4 before and after destriping for analysis. Fig. 19 shows the profiles along line 150 in the original and corrected images of different methods. The horizontal axis represents the sample location in the image, where the vertical axis is the response DNs. The profile in the original image fluctuates because of the stripe, as shown in Fig. 19(a). Therefore, the profile in the corrected image should exhibit a somewhat smoother curve. In Fig. 19(d)–(f), we can observe some stripe features in the curves, indicating that the stripe cannot be removed completely. The images corresponding to Fig. 19(c) are relatively smooth. However, there are some mild burrs in curves compared with Fig. 19(b) and (g) and some residual stripes in its results. From the perspective of the distribution of DNs after correction, the MM and our proposed MAM achieve excellent performance in the MODIS image for band 4.

B. Improvement Factor and MRD

Moreover, the improvement factor (IF) [74] and the index mean relative deviation (MRD) [36] are also evaluated. The IF is defined as follows:

$$\begin{aligned} d_R[j] &= \text{Mean}_{\text{IR}}[j] - \text{Mean}_{\text{I}}[j] \\ d_E[j] &= \text{Mean}_{\text{IE}}[j] - \text{Mean}_{\text{I}}[j] \\ \text{IF} &= 10\log_{10}\left(\frac{\sum_j d_R^2[j]}{\sum_j d_E^2[j]}\right) \end{aligned} \tag{7}$$

where j is the column number of a pushbroom image and $\text{Mean}_{\text{IR}}[j]$ and $\text{Mean}_{\text{IE}}[j]$ refer to the mean radiance of the crosstrack profile of the striped and corrected image, respectively. $\text{Mean}_{\text{I}}[j]$ is the mean radiance of the original image. The IF of the six methods on the urban HYDICE image is shown in Table II, in which the higher IF indicates the better improvement of the radiometric quality of the image.

The index MRD measures the distortion of the original images caused by the adopted destriping method. It is defined as

$$\text{MRD} = \frac{1}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n}\frac{|y(i,j) - x(i,j)|}{x(i,j)} \tag{8}$$

where $x(i, j)$ and $y(i, j)$ stand for the pixel value in position (i, j) of the original image and the corrected image, respectively, and m and n are the number of rows and columns in the image, respectively. MRD calculates the average relative differences of the image before and after destriping. Table III shows the MRD results of the six methods on the four images used in our experiments. The lower the MRD values, the stronger ability of the destriping algorithm to retain the original image content.

We expect to improve the radiation quality of the corrected image and reduce image distortion, that is, higher IF and lower MRD. However, it is meaningless to observe the values of the evaluation indicators without considering the destriping effect. In particular, an image with residual stripe will have a lower MRD. Therefore, we only compare the evaluation indicators

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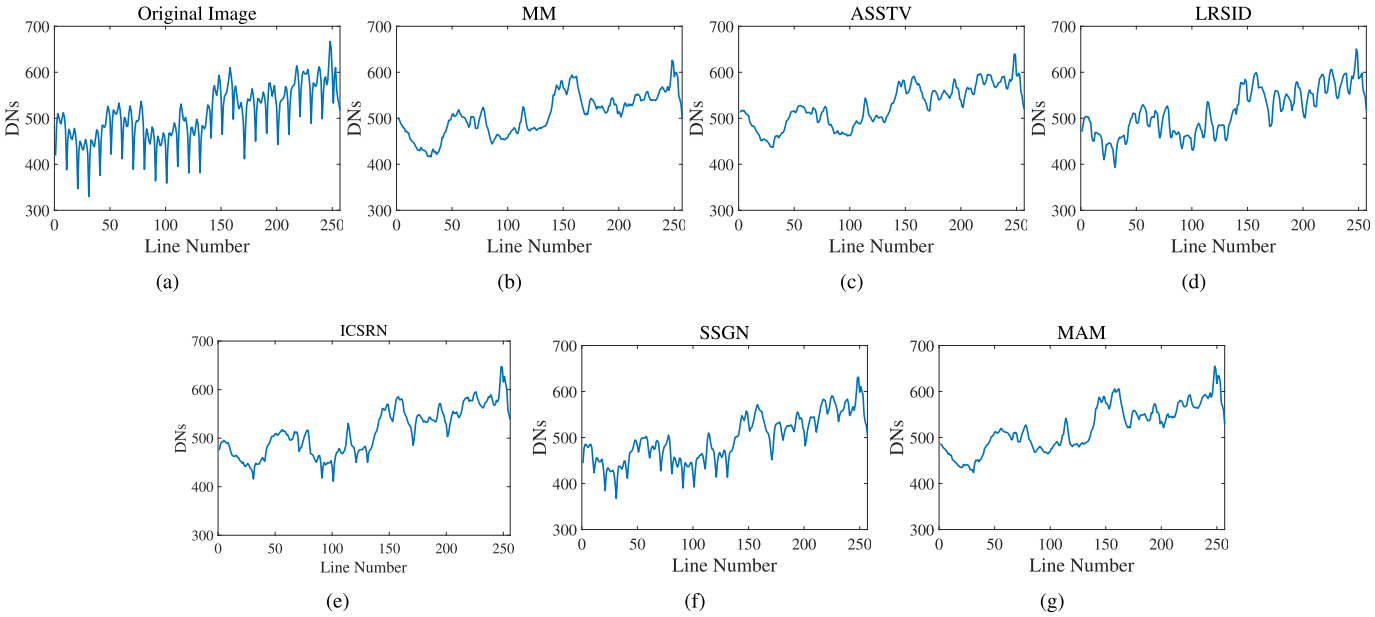


Fig. 19. Profiles along line 150 in band 4 of the MODIS image. (a) Original. (b) MM. (c) ASSTV. (d) LRSID. (e) ICSRN. (f) SSGN. (g) MAM.

TABLE II
IF AND MRD OF THE THREE IMAGES DESTRIPING RESULTS

Method	MM		ASSTV		LRSID		ICSRN		SSGN		MAM	
	Band18	Band29	Band18	Band29	Band18	Band29	Band18	Band29	Band18	Band29	Band18	Band29
IF	0.4584	0.8449	0.4117	1.1856	0.2871	1.8242	1.2793	0.9933	0.4472	0.7360	0.5625	1.3872
MRD	0.3181	0.2494	0.3348	0.2251	0.2844	0.2088	0.3175	0.2306	0.1783	0.1449	0.3029	0.2422

of the methods, which can effectively remove the stripe in the visual effects for subsequent analysis.

From Table II, the proposed method achieves the best IF compared to the MM, ASSTV, and LRSID methods on the Urban image for band 18 and is only lower than LRSID on band 29, which indicates that MAM can improve radiometric accuracy and image quality. In Table III, we can observe that the proposed MAM has the lowest MRD on VNIR and MODIS images, indicating that the distortion of useful information in the original image by MAM during image correction is small. On the Urban image, the MRD of the proposed MAM is only slightly higher than that of the LRSID. Optimization-based methods, including ASSTV and LRSID, show advantages on SWIR images (with the continuous thin stripe) and have smaller MRDs. This is acceptable since we have designed a limit on the difference from the original image when selecting the correction column in the image correction process. The MRD of MAM is not very high compared to other algorithms.

In general, our proposed model has both high IF and low MRD in most cases, which indicates that it is capable of constraining the distortion of the original image while removing the stripe effectively.

C. Quantitative Evaluation on the Whole Simulated Dataset

In addition to the IF and MRD, we also provide four indicators on the whole simulated dataset, including MSNR, MPSNR, MSSIM, and MSAD, to evaluate the image quality and the degree of spectral distortion after destriping. The results are shown in Table IV. We can observe that MAM achieves the best MSSIM and relatively small MSAD, indicating that the structural information of the original image can be better preserved, and spectral distortion is limited. The proposed model has comprehensively comparable results in the other two evaluation indicators.

Although our method is slightly inferior to LRSID in the simulated data according to the MSNR and MPSNR, it performs much better on the real images. The proposed method can effectively remove various stripes, especially as shown in Figs. 11(d) and (g), 12(d) and (g), 15(d) and (g), and 16(d) and (g).

D. Image Uniformity

The uniformity of the image can be improved after destriping. Therefore, the destriping ability of algorithms is quantitatively validated by comparing the nonuniformity of images before and after correction. Two uniform regions with the size of 20×60 marked by the red box in Fig. 20 are selected from SWIR images.

The nonuniformity of an image can be calculated by referring to [75]

$$U = \frac{1}{\bar{S}} \times \sqrt{\frac{\sum_{i=1}^{N_s} (S_i - \bar{S})^2}{N_s}} = \frac{\sigma_i}{\bar{S}} \tag{9}$$

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TABLE III
MRD AND CHANGES IN BEFORE AND AFTER CORRECTION RESULTS

Image		Method	MRD	MV	Change in MV
Urban	Band 18	Striped image	/	136.58	/
		MM	0.3181	139.97	0.0248
		ASSTV	0.3348	154.23	0.1293
		LRSID	0.2844	143.21	0.0486
		ICSRN	0.3175	154.44	0.1307
		SSGN	0.1783	136.77	0.0014
		MAM	0.3029	139.70	0.0229
	Band 29	Striped image	/	155.37	/
		MM	0.2494	156.86	0.0096
		ASSTV	0.2251	156.88	0.0097
		LRSID	0.2088	153.04	0.0150
		ICSRN	0.2306	170.95	0.1003
		SSGN	0.1449	157.37	0.0148
		MAM	0.2422	154.00	0.0088
VNIR	Band 9	Striped image	/	10153	/
		MM	0.1329	10271	0.0117
		ASSTV	0.0794	10339	0.0184
		LRSID	0.0152	10132	0.0020
		ICSRN	0.0297	9822	0.0325
		SSGN	0.0227	10266	0.0112
		MAM	0.0313	10087	0.0065
	Band 59	Striped image	/	4394	/
		MM	0.0595	4429	0.0080
		ASSTV	0.0517	4172	0.0505
		LRSID	0.0222	4301	0.0210
		ICSRN	0.0280	4281	0.0256
		SSGN	0.0204	4433	0.0089
		MAM	0.0463	4407	0.0030
SWIR	Band 250	Striped image	/	16373	/
		MM	0.1230	16004	0.0225
		ASSTV	0.0711	16123	0.0153
		LRSID	0.0718	15844	0.0323
		ICSRN	0.0794	16930	0.0340
		SSGN	0.0659	15817	0.0340
		MAM	0.1206	16209	0.0100
	Band 314	Striped image	/	21833	/
		MM	0.1142	21547	0.0131
		ASSTV	0.0694	22849	0.0465
		LRSID	0.0570	22355	0.0239
		ICSRN	0.0582	21987	0.0070
		SSGN	0.0847	23321	0.0681
		MAM	0.1080	21651	0.0084
MODIS	Band 4	Striped image	/	523.28	/
		MM	0.0501	526.97	0.0071
		ASSTV	0.0462	533.50	0.0195
		LRSID	0.0332	527.84	0.0087
		ICSRN	0.0369	528.71	0.0104
		SSGN	0.0370	514.19	0.0174
		MAM	0.0482	537.40	0.0270
	Band 6	Striped image	/	513.85	/
		MM	0.0445	515.18	0.0026
		ASSTV	0.0462	502.85	0.0195
		LRSID	0.0474	498.14	0.0306
		ICSRN	0.0271	512.90	0.0019
		SSGN	0.0266	518.59	0.0092
		MAM	0.0413	504.89	0.0174

where U represents the image nonuniformity, $\bar{S}$ is the mean of the DNs of the signal, N_s is the sample number, S_i is the DN of pixel i, and σ_i stands for the standard deviation of pixel i.

TABLE IV
COMPARISON OF THE MVS OF FOUR QUALITY METRICS ON THE WHOLE URBAN DATASET

Methods	MSNR	MPSNR	MSSIM	MSAD
Stripe	13.18	25.97	0.73	0.22
MM	11.67	23.82	0.88	0.15
ASSTV	15.40	28.30	0.90	0.18
LRSID	16.58	29.47	0.90	0.15
ICSRN	13.98	26.38	0.88	0.18
SSGN	15.56	27.18	0.91	0.16
MAM	15.89	27.91	0.91	0.16

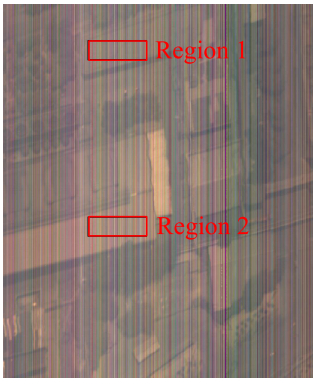


Fig. 20. Two ground target regions in the SWIR image.

TABLE V
IMAGE NONUNIFORMITY USING SIX METHODS

Method	Band 250		Band 314	
	Region 1	Region 2	Region 1	Region 2
Original image	0.0912	0.0648	0.0829	0.0514
MM	0.0375	0.0280	0.0573	0.0372
ASSTV	0.0466	0.0245	0.0523	0.0131
LRSID	0.0525	0.0277	0.0533	0.0185
ICSRN	0.0520	0.0289	0.0584	0.0175
SSGN	0.0423	0.0445	0.0731	0.0488
MAM	0.0366	0.0114	0.0525	0.0153

The calculated results of six methods are shown in Table V. It can be seen from Table V that MAM has the most significant improvements in image uniformity/smoothness on band 250. On band 314, the results of MAM are only slightly higher than those of ASSTV, indicating that the TV model improves the uniformity of images. Compared with other destriping methods, MAM considers the smaller horizontal same material gradient when selecting the optimal correction column from the correction candidates, making the corrected image more uniform.

E. Image Information Preservation

Compared with general image processing, hyperspectral image destriping increases the image clarity, preserves the true information, and improves the quantitative accuracy. To evaluate the information retention capacity of MAM, the MV of

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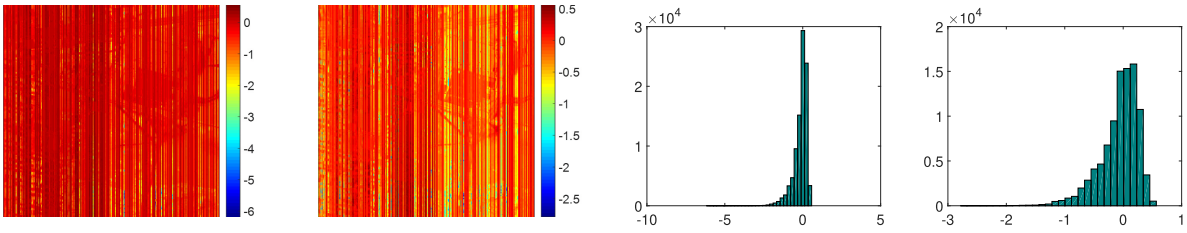


Fig. 21. Heatmaps and histograms of change ratio for Urban image for bands 18 and 29. (a) Heatmap for band 18 and (c) corresponding histogram. (b) Heatmap for band 29 and (d) corresponding histogram.

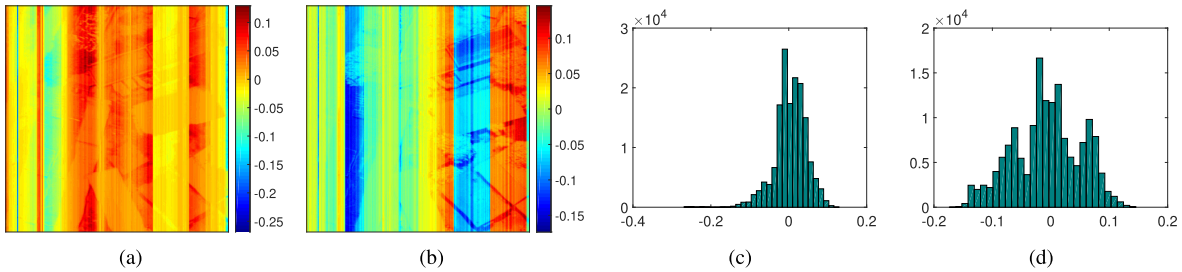


Fig. 22. Heatmaps and histograms of change ratio for VNIR image for bands 9 and 59. (a) Heatmap for band 9 and (c) corresponding histogram. (b) Heatmap for band 59 and (d) corresponding histogram.

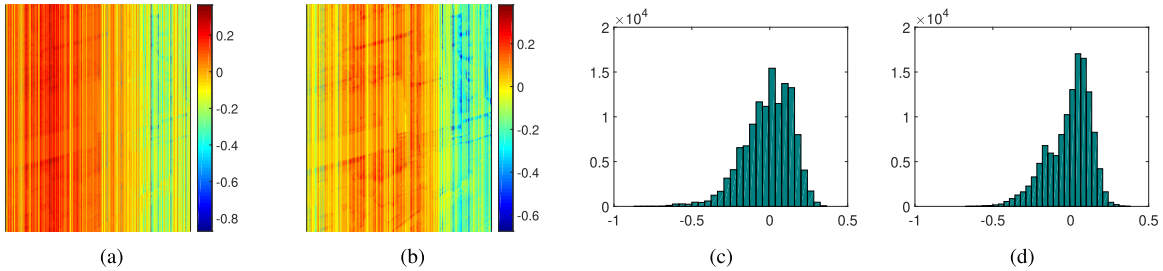


Fig. 23. Heatmaps and histograms of change ratio for SWIR image for bands 250 and 314. (a) Heatmap for band 250 and (c) corresponding histogram. (b) Heatmap for band 314 and (d) corresponding histogram.

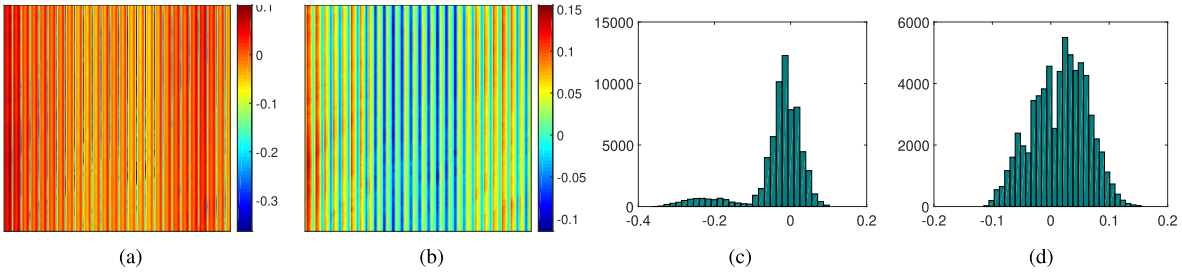


Fig. 24. Heatmaps and histograms of change ratio for MODIS image for bands 4 and 6. (a) Heatmap for band 4 and (c) corresponding histogram. (b) Heatmap for band 6 and (d) corresponding histogram.

each image is calculated by referring to

$$\mathrm{MV} = \frac{\sum_{i=1}^{N_s}(S_i)}{N_s} \tag{10}$$

where MV is the mean of the DNs of the signal, N_s is the sample number, and S_i is the DN of pixel i. The change in MV reflects the degree of information preservation before and after destriping. The results of changes in MVs are listed in Table III. As shown in Figs. 21–24, we use the heatmaps and histograms to show the change ratio of each pixel after correction from the perspective of spatial distribution and statistics, respectively. The change ratios of pixels indicate that the difference between the original image and the corrected image is divided by the original image.

The following conclusions can be drawn from Table III and Figs. 21–24.

1) We observe some particular cases where some methods have a relatively low change value, such as ICSRN and SSGN in Urban and MODIS images. Because the evaluation indicators' values without considering the destriping effect are meaningless, we should compare the evaluation indicators of the methods that can

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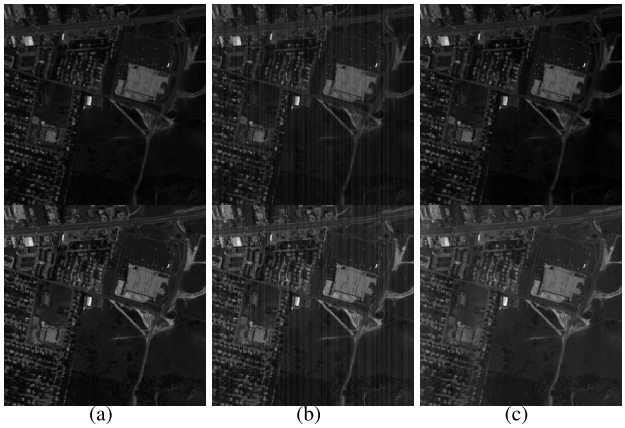


Fig. 25. Corrected results of our MAM on Urban with additive stripe. (a) Original image. (b) Striped image. (c) Corrected image. Bands 7 and 37 are displayed sequentially from top to bottom.

effectively remove the stripe in the visual effects for subsequent analysis. In general, the proposed MAM obtains the least MV change on the Urban image for band 29, VNIR image, and SWIR image. Others, such as the Urban image for band 18 and MODIS image, and MAM are only slightly higher than LRSID and MM, respectively. The above indicates that the adaptive MAM preserves the original information to the most considerable extent.

2) The heatmaps in Figs. 21(a) and (b)–24(a) and (b) represent the change ratio of each pixel after the MAM destriping in the image. It can be observed that heatmaps have striped feature distribution, in which the distribution is consistent with the stripe noise. The small pixel change ratio corresponds to the lighter interference from the stripe in the original image in the spatial position and vice versa. In Figs. 21(a) and (b)–24(a) and (b), we can very intuitively observe that most change ratios are near 0.
3) The histograms in Figs. 21(c) and (d)–24(c) and (d) further verify the results shown in Table III, which shows the statistics of change ratios for pixels after destriping. As shown in Figs. 21–24, the change ratios of most pixels are relatively small, indicating that a large amount of original information has been reserved.

F. Validation on Additive Stripe Noise

To evaluate the destriping performance of different stripes, we conduct additional experiments of our method on the Urban image with additive stripe. The additive stripe generation is formulated as: $V'_{l,j}(\varphi) = v_{l,j}(\varphi) + C_{l,j}$, where $C_{l,j}$ is the additive noise values. The same column shares the same $C_{l,j}$. The results are shown in Fig. 25. It can be observed that our MAM achieves advanced destriping performance on the additive stripe, indicating that the proposed MAM and correction methods are flexible in handling diverse stripes.

G. Large-Swath Image Stripe Removal

Our MAM can also be applied to large-swath remote sensing images, such as MODIS image with 1000 $times$ 1000.

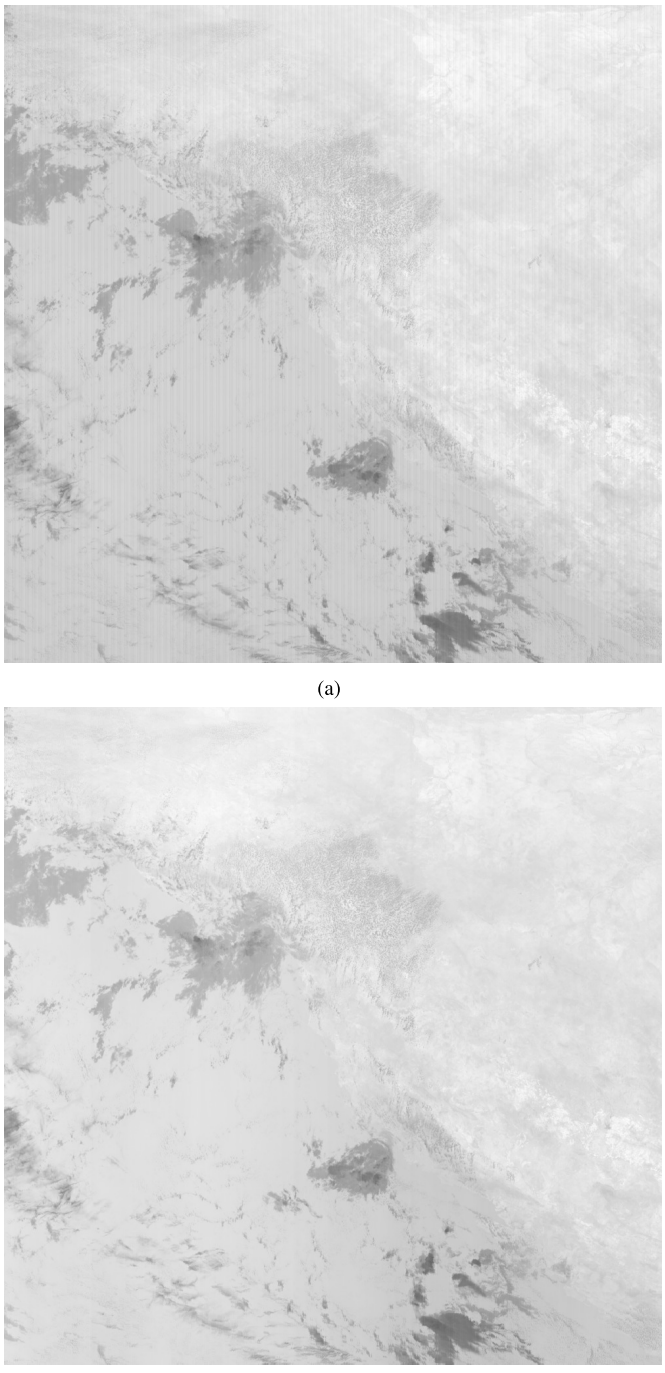


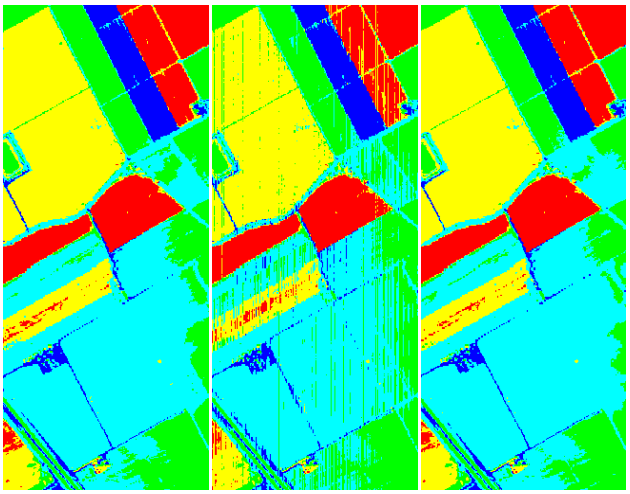
Fig. 26. Real MODIS band 10 image with 1000 × 1000 size. (a) Original image. (b) Corrected image.

The result is shown in Fig. 26. We can observe a lot of stripe noise in the original image from Fig. 26(a). The proposed MAM can remove the large-swath image stripe, as shown in Fig. 26(b), and the image structure information has not been changed. Since the MAM corrects the image by determining a more accurate MAM, it can also achieve excellent destriping effects on the larger images.

H. Effectiveness for High-Level Task

Destriping can effectively improve the visual quality of images and the effect of the subsequent application. Therefore,

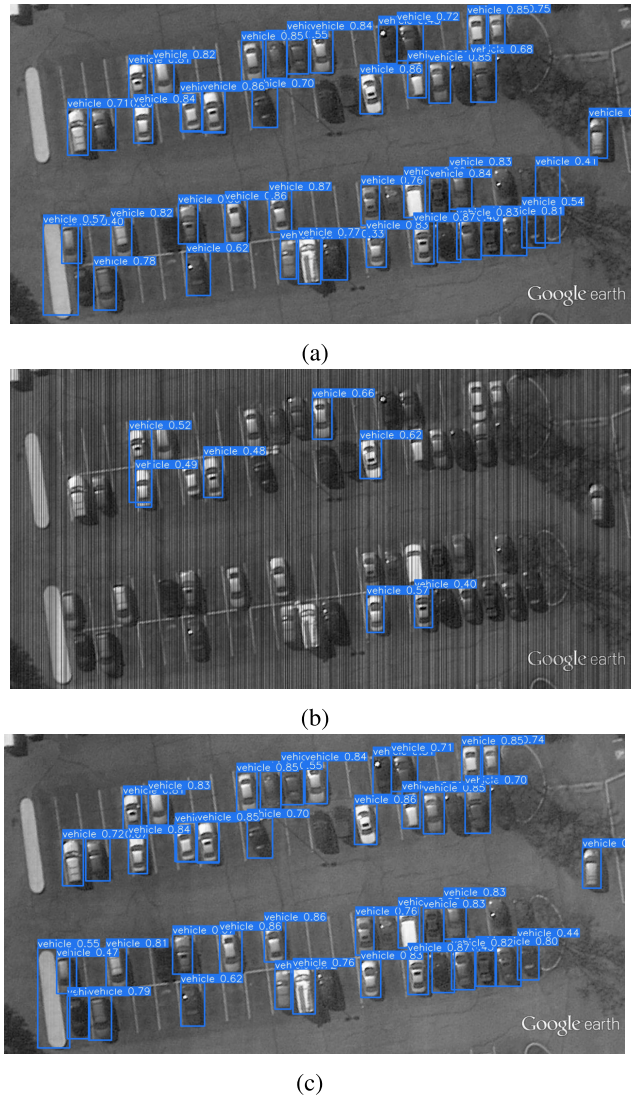
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(a) (b) (c)

Fig. 27. Evaluation of destriping results of the proposed MAM on segmentation task. Segmentation results for (a) original image, (b) striped image, and (c) corrected image.

we further demonstrate the effectiveness of the MAM on high-level task, including segmentation and object detection. The unsupervised K-means is adopted as semantic segmentation for the original, striped, and corrected images of Salinas. The number of classes is five, and the maximum iteration is set as ten times. The segmentation results are shown in Fig. 27. From Fig. 27(b), we can see that the segmentation result is severely affected by the stripe. The stripe seriously impairs the spectral information of different ground objects, leading to mistakes during the clustering process. Most of the segmentation edges are along with the stripe instead of the boundaries of ground objects. After destriping, the proposed model significantly corrects the spectral data. As shown in Fig. 27(c), the segmentation result has also been improved and is similar to the original image. For object detection, we take one UCAS-High Resolution Aerial Object Detection (UCAS-AOD) image as an example, and YOLO-V5 is employed for object detection. The results are shown in Fig. 28. The stripe severely damages the appearance feature of the vehicles, resulting in that only the seven vehicles are detected. After destriping by our method, the object detection performance is similar to the original image. Thus, the proposed MAM could promote the performance of the subsequent high-level task.



(a)

(b)

(c)

Fig. 28. Evaluation of destriping results of the proposed MAM on object detection task. Detection results for (a) original image, (b) striped image, and (c) corrected image.

I. Running Time

To measure the efficiency of the proposed method, we show the running time of six compared methods on the different data experiments in Table VI, in which MM, ASSTV, LRSID, and MAM run in MATLAB 2016a on the same PC with an Intel i5-10210U CPU at 1.6-GHz and 8-GB memory. The running time of ICSRN and SSGN is obtained in MATLAB 2016a, Caffe, and pytorch 1.1.0 on the same workstation with an Intel i7-8700k CPU at 3.8-GHz and 128-GB memory and NVIDIA GTX1080Ti, respectively. We can observe from Table VI that the running time of MAM is not the fastest compared to other algorithms, especially MM and deep learning methods. A large amount of processing time of MAM is spent for MAM and the generation and selection of optimal correction column.

TABLE VI

RUNNING TIME (IN SECONDS) OF THE METHODS IN THE DIFFERENT DATA (FOR ONE BAND)

Methods	Urban	VNIR	SWIR	MODIS
MM	0.01	0.01	0.01	0.01
SSGN	0.07	0.10	0.07	0.06
ICSRN	0.21	0.35	0.26	0.15
ASSTV	1.22	1.01	3.93	0.56
LRSID	14.81	14.44	44.47	6.54
MAM	23.10	25.24	51.01	9.45

Although our method is not the fastest, we perform better than the others mentioned above. As shown in Figs. 11–16, MAM can effectively remove all kinds of stripes compared to other algorithms, especially the wide stripe (Figs. 11 and 12). Furthermore, MAM achieves relatively excellent results no

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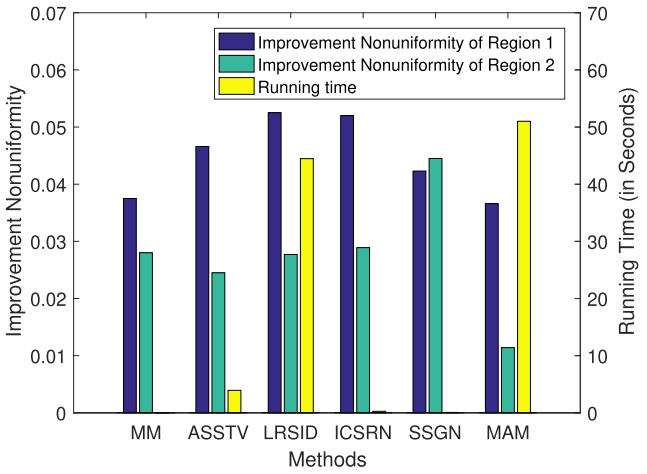


Fig. 29. Comparison of running time and image uniformity improvement for six methods.

matter the degree of image distortion or preserving the original image information. Fig. 29 shows the comparison of the uniformity improvement (the reported value is the smaller the better) and the running time of the six algorithms on the 250 band of SWIR image. We can observe that although the running time of MAM is not fast, it possesses the best improvement on the image nonuniformity.

VI. Conclusion

In this article, we propose an adaptively MAM method for hyperspectral image destriping without making any explicit assumptions of data. The MAM method is designed to search the same material pixels in a local region by the PMA and stripe feature distance, to calculate the correction coefficients for correcting the image column by column. Moreover, to improve the robustness of the destriping process, we propose to generate multiple correction candidates for each column and select the optimal correction column by considering both image uniformity and content preservation. We compare the MAM method with five classic destriping methods on both synthetic and real images. The superior experimental results indicate that the proposed method can effectively improve the uniformity and preserve the hyperspectral image structure while removing the stripe noise. Since our proposed method corrects the image columnwisely based on the MAM, the destriping effect may be somewhat limited when the region of the same material in the given scene is too small to match the corrected pixels in two adjacent columns. We will conduct further investigations in our future work regarding the complex material distributions.

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